TEXORA · LECTURE SHEET

Polymerization Techniques

The two great mechanisms — chain-growth and step-growth — and the industrial reactor techniques (bulk, solution, suspension, emulsion, melt and interfacial) by which monomers are actually turned into polymer.

TE-2107 · POLYMER SCIENCE AND ENGINEERING · TOPIC 2 · LEVEL 2 TERM 1

1. Two mechanisms, two growth patterns

1.1 Chain-growth (addition) polymerisation

An **initiator** creates an active centre — a free radical, cation or anion — on one monomer. That centre adds monomer after monomer in rapid succession: the chain grows to full length in a fraction of a second, then is terminated. At any moment the reactor holds monomer and finished polymer, but almost no half-grown chains. The three stages of the commonest version, **free-radical polymerisation**:

1. **Initiation.** An initiator such as benzoyl peroxide or AIBN decomposes under heat or light into free radicals; a radical adds to the C=C bond of a monomer, creating a new, larger radical.
2. **Propagation.** The chain radical adds further monomers, hundreds or thousands per second, the radical always sitting at the growing tip.
3. **Termination.** Two growing radicals meet and combine (or disproportionate), killing both chains. High molar mass is reached from the very start of the reaction, and conversion mainly increases the *number* of chains.

Monomers: ethylene, propylene, vinyl chloride, styrene, acrylonitrile, methyl methacrylate — all C=C compounds. No by-product is formed; the repeat unit equals the monomer.

1.2 Step-growth (condensation) polymerisation

Monomers carrying two reactive functional groups (–OH, –COOH, $–NH_2$) react with each other in ordinary chemical steps: monomer + monomer → dimer, dimer + monomer → trimer, dimer + dimer → tetramer, and so on. Any two molecules in the pot can react, so chains grow slowly and uniformly, and — critically — high DP is reached only when the reaction is driven almost to completion. A small molecule, usually water or methanol, is eliminated at each bond formed. Because the by-product stops the reaction reaching completion if it accumulates, industrial step-growth processes must remove it continuously (vacuum, inert-gas sweep, thin-film reactors).

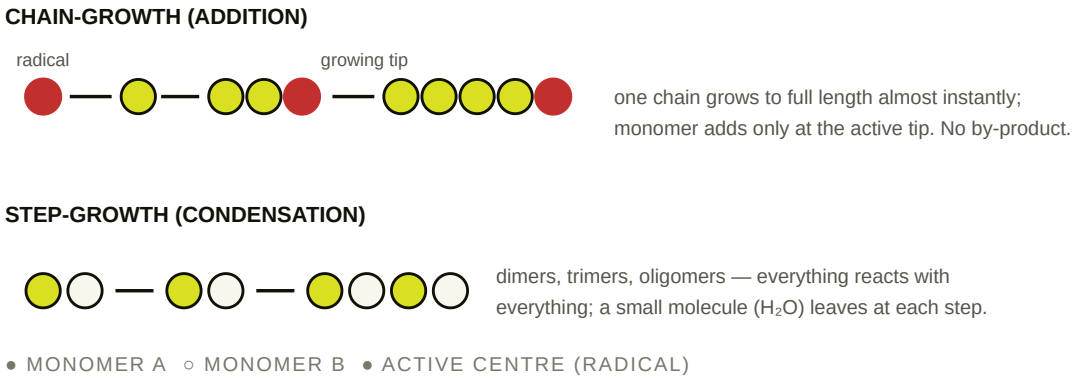


Figure 1 — The two growth patterns. Chain-growth: monomer adds only at an active tip, chain by chain. Step-growth: all species grow together, one bond at a time, shedding water as they go.

2. The Carothers equation — why step-growth demands near-total conversion

For a step-growth polymerisation of exactly balanced bifunctional monomers, Carothers showed that the number-average DP depends only on the fraction p of functional groups that have reacted:

$$X_n = 1 \div (1 - p)$$

WORKED EXAMPLE — CAROTHERS EQUATION

p = 0.98 → X_n = 1 ÷ 0.02 = **50** (still an oligomer).

p = 0.99 → X_n = **100** (minimum useful fibre DP).

p = 0.995 → X_n = **200**; p = 0.999 → X_n = **1000**.

So the last 1 % of conversion does more than the first 98 %: this is why polyester and nylon plants fight for the final fraction of a percent — pure monomers, exact stoichiometric balance, and continuous removal of water under vacuum. A 1 % excess of one monomer is equally fatal, because every chain then ends with the excess group and cannot grow further.

Ring-opening polymerisation is a third route sitting between the two: a cyclic monomer (caprolactam → nylon 6; ethylene oxide) opens and adds to a growing chain without eliminating anything, giving step-growth-like structures by a chain mechanism.

3. Industrial techniques (reactor methods)

3.1 Bulk (mass) polymerisation

Monomer alone, plus initiator — nothing else. The product is the purest possible, and the method is used for polystyrene sheet and PMMA casting. Its problem is the reaction itself: polymerisation is

exothermic, and as polymer forms the mixture becomes viscous, traps heat, and can run away (the Trommsdorff or gel effect accelerates the reaction further). Industrial use is therefore limited to partial conversion or specially cooled reactors; PET and nylon are made in stirred melt reactors that face the same heat- and mass-transfer battle.

3.2 Solution polymerisation

Monomer and initiator are dissolved in a solvent. The solvent dilutes the reaction, carries away heat and keeps viscosity manageable — control is easy, but the solvent must later be removed and recovered, and the polymer's molar mass is lower (chain transfer to solvent). The technique survives in textiles because acrylic fibre plants polymerise in solution (e.g. in DMF or aqueous sodium thiocyanate) and the *same solution* goes straight to the spinning bath as dope — polymerisation and spinning in one continuous line.

3.3 Suspension polymerisation

Water-insoluble monomer is broken into droplets (0.05–1 mm) in water by vigorous stirring, with a suspending agent (e.g. polyvinyl alcohol) preventing coalescence. Each droplet is a tiny bulk reactor; the heat flows easily into the water. The product comes out as beads that are filtered, washed and dried. PVC for coatings and some polystyrene are made this way.

3.4 Emulsion polymerisation

Monomer is emulsified in water with a surfactant (soap), forming micelles. A water-soluble initiator starts chains inside the micelles, which grow into polymer particles of 0.05–0.2 µm — a **latex**. Heat control is excellent, molar mass is very high, and the latex can be used directly (textile coatings, carpet backing, adhesive finishes). Synthetic rubbers (SBR) and acrylic emulsions for textile finishing are made this way; the penalty is surfactant contamination of the polymer.

TECHNIQUE	MEDIUM	PRODUCT FORM	MAIN ADVANTAGE	MAIN DISADVANTAGE
Bulk	None	Solid mass / melt	Purest polymer	Heat removal very difficult
Solution	Organic solvent	Polymer solution (dope)	Good control; dope can be spun directly	Solvent recovery; lower molar mass
Suspension	Water	Beads	Easy heat removal, easy isolation	Needs washing/drying
Emulsion	Water + surfactant	Latex (fine dispersion)	Highest molar mass, direct use as latex	Surfactant residue

3.5 Techniques specific to step-growth

- **Melt polycondensation** — monomers are heated above the polymer's melting point (PET: about 270–285 °C) and water/glycol is pulled off under vacuum in the final stage. The standard route to polyester and nylon salts.
- **Interfacial polycondensation** — two monomers dissolved in two immiscible liquids (e.g. a diamine in water, a diacid chloride in an organic solvent) react instantly at the liquid–liquid interface; a film of polymer can be drawn continuously from the boundary (the classroom "nylon rope trick"). Used for speciality polyamides and polycarbonate-type products where heat-sensitive monomers rule out the melt route.
- **Solution polycondensation** — both monomers in one solvent at modest temperature, for heat-sensitive high-performance polymers such as aramids.
- **Solid-state polymerisation (SSP)** — polyester chips are heated below their melting point under vacuum or nitrogen so end groups keep reacting; used to raise bottle-grade and industrial-yarn PET to very high molar mass without melt degradation.

Glossary

Initiator — a compound that decomposes into radicals (or ions) to start chain growth. **Propagation** — repeated monomer addition at the active tip. **Termination** — destruction of active centres, ending chain growth. **Conversion (p)** — the fraction of functional groups (or monomer) that has reacted. **Latex** — a stable aqueous dispersion of polymer particles from emulsion polymerisation. **Dope** — the polymer solution prepared for wet or dry spinning. **Chain transfer** — the active centre jumping to another molecule, ending one chain and starting another.

Key points

• Chain-growth: full-length chains from the start, growth only at active tips, no by-product. Step-growth: everything reacts with everything, DP explodes only near 100 % conversion.

• Carothers: $X_n = 1/(1-p)$ — p = 0.99 gives DP 100; stoichiometry and by-product removal are decisive.

• Bulk gives purity but dangerous heat build-up; solution gives control and spinnable dope; suspension gives beads; emulsion gives latex and the highest molar mass.

• Step-growth industrially: melt polycondensation for PET/nylon, interfacial for heat-sensitive monomers, solid-state finishing for very high DP polyester.

Practice

1. Using the Carothers equation, what conversion is needed for X_n = 250? Comment on what this demands of monomer purity.
2. A reactor contains only monomer and full-length polymer, with virtually no chains of intermediate length. Which mechanism is operating, and why does the mixture look like this?
3. An acrylic plant polymerises in DMF solution and pipes the product straight to spinning. Which technique is this, and what two costs does it avoid compared with bulk polymerisation followed by separate dope preparation?
4. Why does emulsion polymerisation give higher molar masses than bulk polymerisation of the same monomer?
5. State two reasons why the water formed in PET melt polycondensation must be removed continuously.

Answer key

1. 250 = 1/(1−p) → 1−p = 0.004 → p = **0.996**, i.e. 99.6 % of all functional groups reacted. Impurities that cap chains or unbalance the stoichiometry at even the 0.5 % level make this unreachable, so monomer purity and ratio control must be better than about 1 part in 250.
2. Chain-growth (addition). Chains are born, grow to full length in moments and die one at a time, so the pot always holds untouched monomer plus finished chains; intermediates exist only as the tiny population of actively growing tips.
3. Solution polymerisation. It avoids (i) isolating, drying and re-dissolving the polymer — no separate dope-making step — and (ii) the heat-removal crisis of bulk polymerisation, since the solvent carries the reaction heat away.
4. In emulsion polymerisation each latex particle typically contains only one growing radical, isolated from radicals in other particles, so termination (which needs two radicals to meet) is rare; chains grow far longer before dying than in bulk, where radicals meet freely.
5. (i) Polycondensation is reversible — accumulated water drives the equilibrium backwards, hydrolysing ester links and capping DP (Carothers). (ii) At melt temperature, water also degrades the polymer by hydrolysis during processing, lowering molar mass and creating carboxyl end groups that worsen dyeability and stability.